Complement system II:

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SGL 5

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Objectives:

- The students must learn all detail about the definition, characteristics, pathways and biological functions of the complement cascade system.

The complement system:-

- **Definition:-**
- The term (complement) means to supplement (the effect of other components of the immune system, eg. antibody).
- **Complement characteristic:-**
- 1-Complement system consists of a series of proteins that work to "complement" the work of antibodies in destroying microorganism.

The complement system pathways:-

- There are three pathways for activation of complement cascade:-

- 1-Classical pathway.
- 2-Alternative pathway.
- 3-Lectin pathway.

The complement system:-

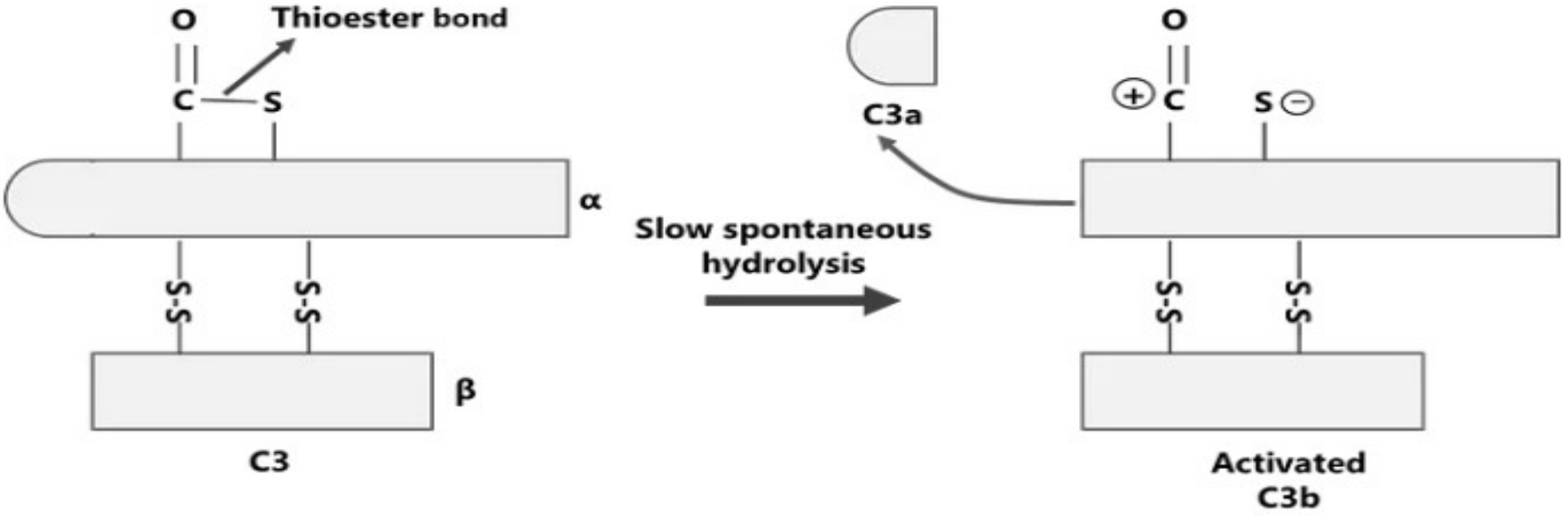
- ***All pathways can be divided into three phases:-***
- **1-Initiation phase: -** This phase consists of recognition events.
- **2-Amplification phase: -** The result of this phase is activation of C3
- **3-Membrane attack phase: -** The result of this phase is target cell lyses.

Alternative pathway

- It **does not need antigen-antibody** complexes to activate this complement system.
- The alternative pathway is considered a component of the **innate** immune system because no Ab is required.
- This complement **activation** pathway involves four proteins- C3, Factor B, Factor D, and properdin.

Alternative pathway:

- Alternative pathway is a primitive defense system.
- This pathway is **initiated** by **cell surface components** that are foreign to the host, such as
- Lipopolysaccharides from **gram-negative** bacteria cell walls
- Teichoic acids from **gram-positive** bacteria cell walls
- Zymosan from **fungal** and **yeas**t cell walls.
- Some viruses, parasites, virus-infected cells, tumor cells, etc., can also activate the alternative pathway of the complement system.
- In this pathway, the thioester bond of serum C3 undergoes slow spontaneous hydrolysis to yield C3a and C3b.



- This C3b component can bind to foreign surface Ags.
- It can even bind on the host's own cells.
- The membrane of mammalian cells has **high levels of sialic acid**, which enables the rapid inactivation of C3b molecules on the host cells.
- Foreign antigenic surfaces have **low sialic acid levels** therefore, C3b bound to them remains active and can activate the complement system.

2- Alternative pathway

- **C3b interacts with factor B** to form C3bB that became susceptible to enzymatic cleavage by **factor D** , two fragment are formed **Ba**(chemotactic for neutrophils) and **Bb** which bound to **C3b** forming **C3bBb** complex(C3 convertase).
- **Properdin** is a protein that stabilizes this complex leading to generate more **C3b**.

2- Alternative pathway

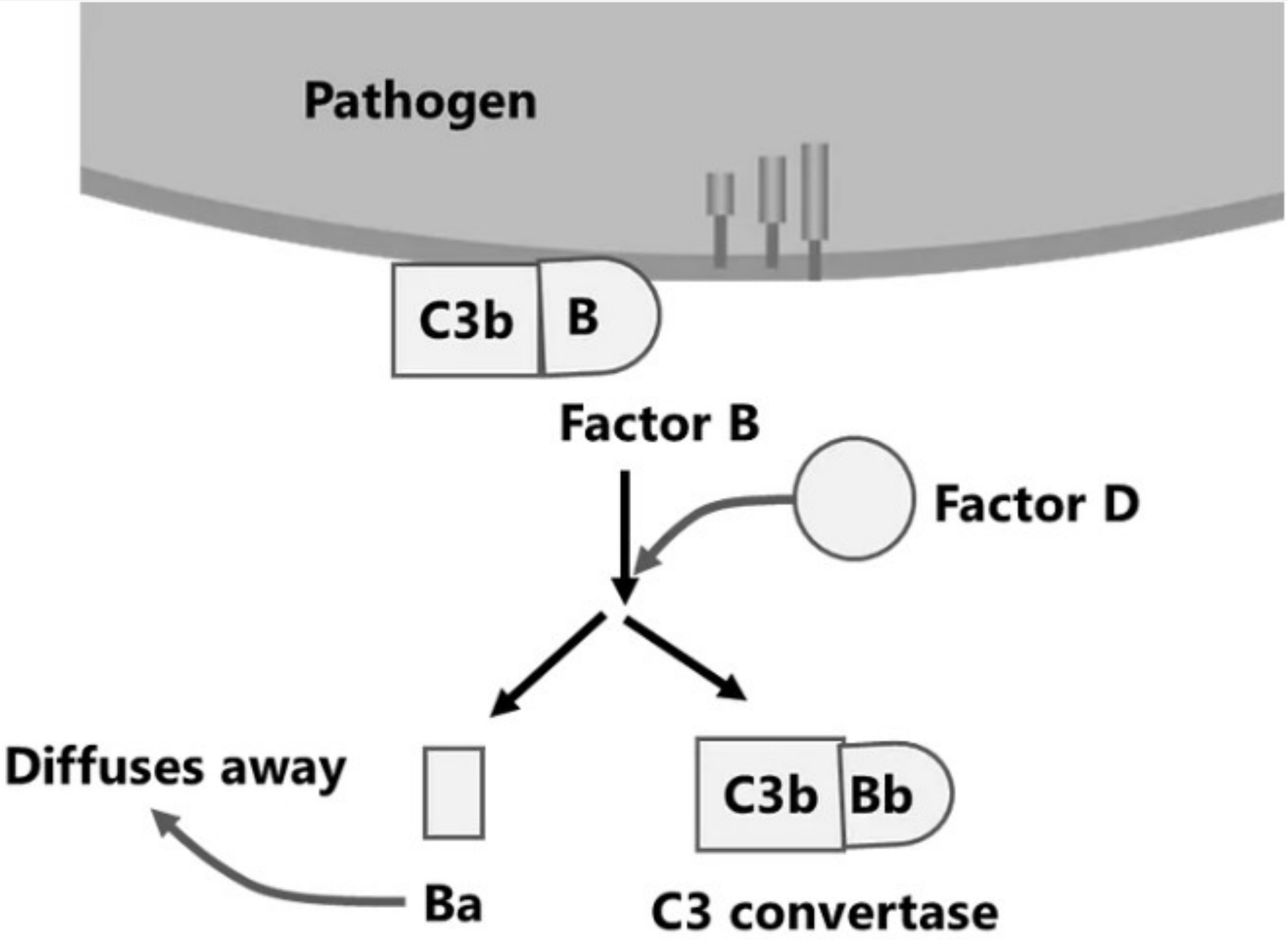
- As more C3b is generated, the complex expands with **numerous C3b** molecules attached to a **single Bb**(**C3bnBb**, where n>1) and becomes a **C5 convertase** that capable of cleaving C5 and thereby initiating the membrane attack pathway.

- Initiation by C3b is prevented by inactivation with **factor H and I**.

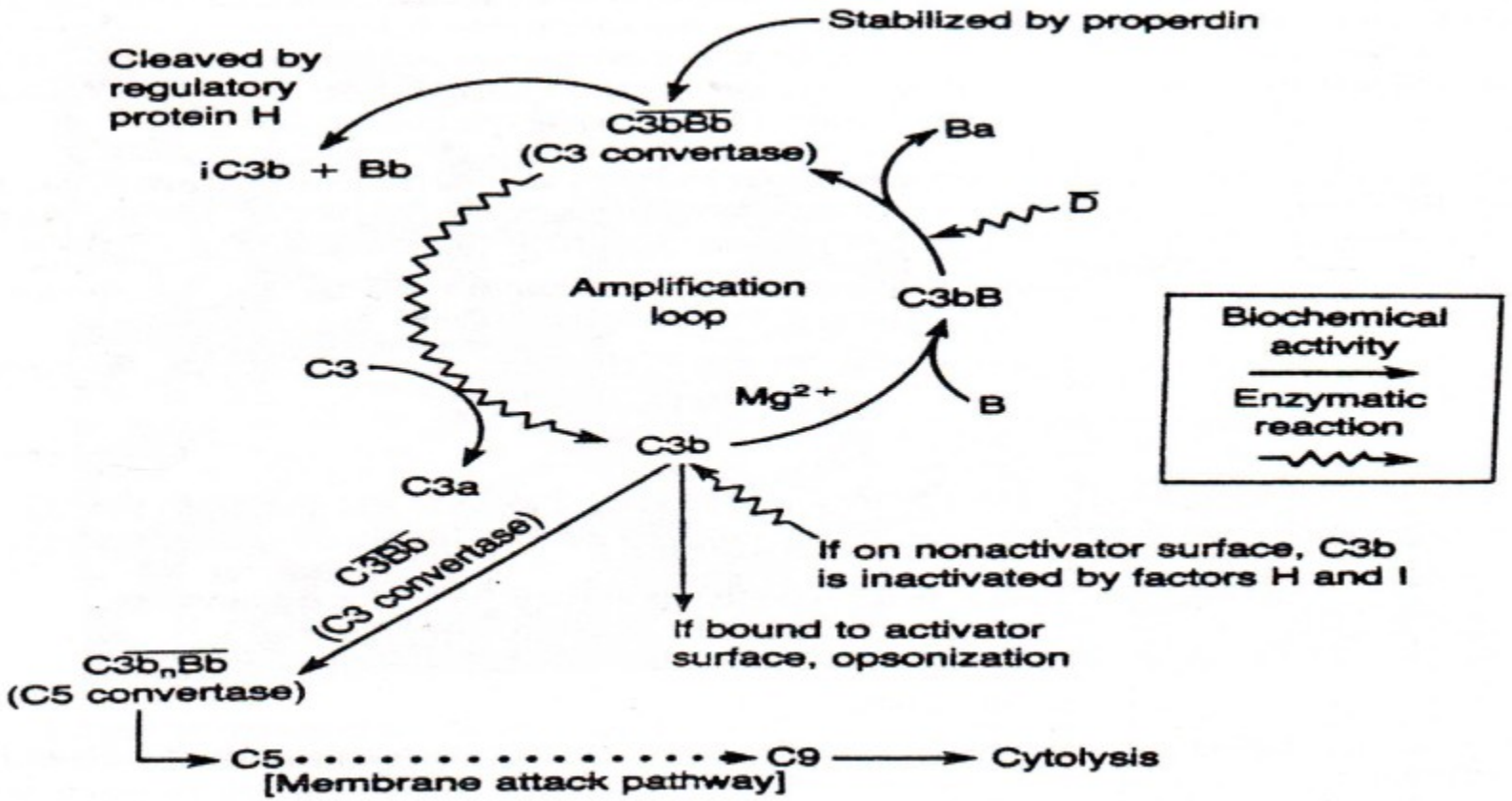
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Pathogen
C3b
B
Factor B
Factor D
Diffuses away
Ba
C3b
Bb
C3 convertase

The Alternative pathway:-



(Hyde, 2000)

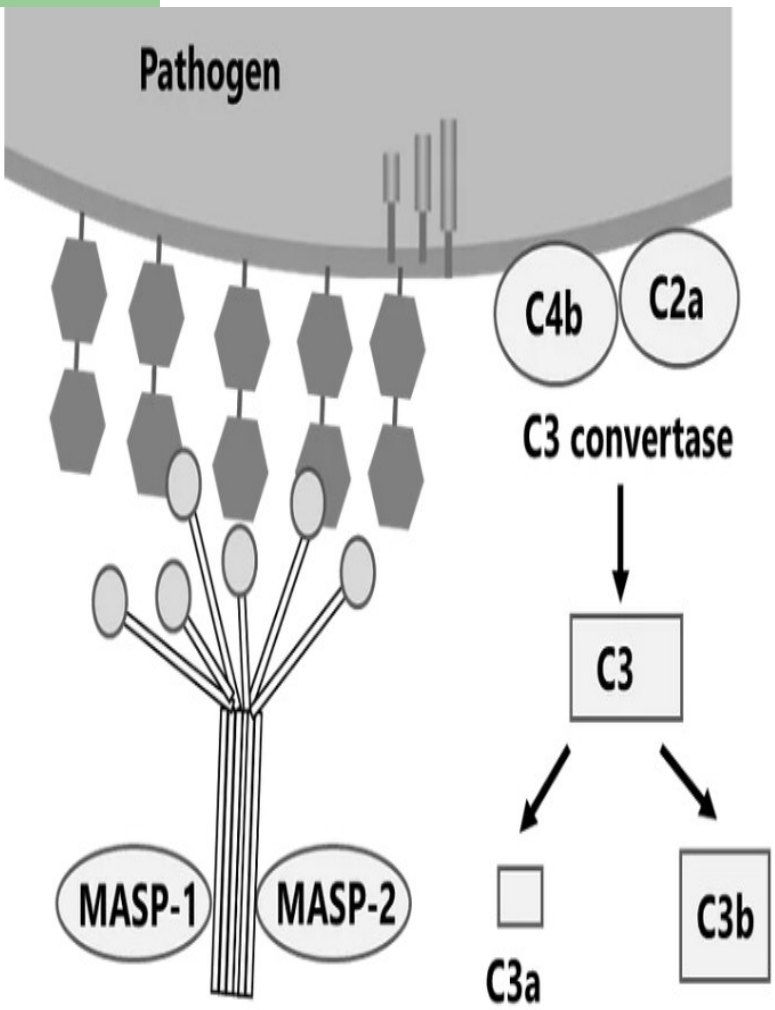
3- Lectin pathway:-

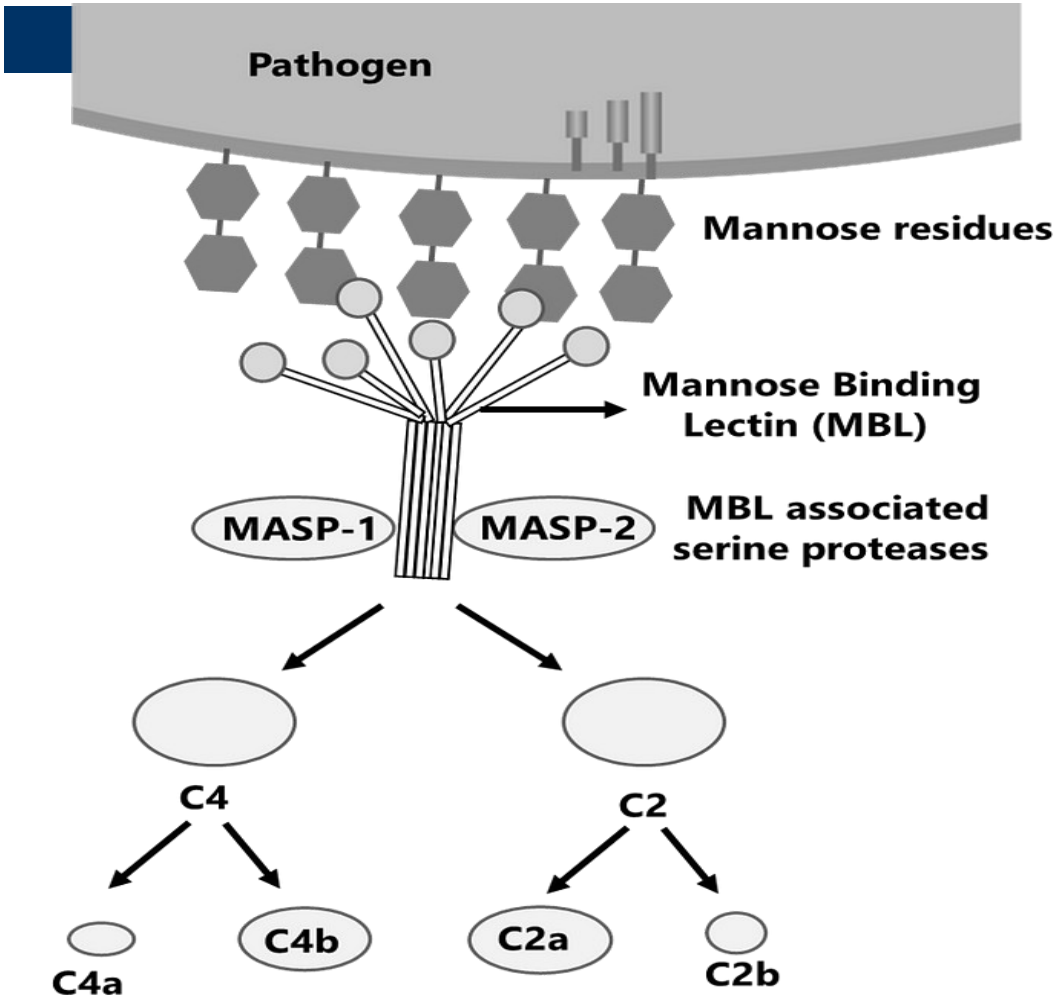
- The third pathway by which the complement system can be activated.
- The lectin pathway does not depend on Ab for its activation.
- This pathway is initiated by the lectins.
- Lectins are the proteins that can recognize and **bind** to the specific carbohydrate residues(**mannose residues)** on the **target cells**.
- This pathway is also named the **MBL pathway** or **Mannan binding lectin pathway**.

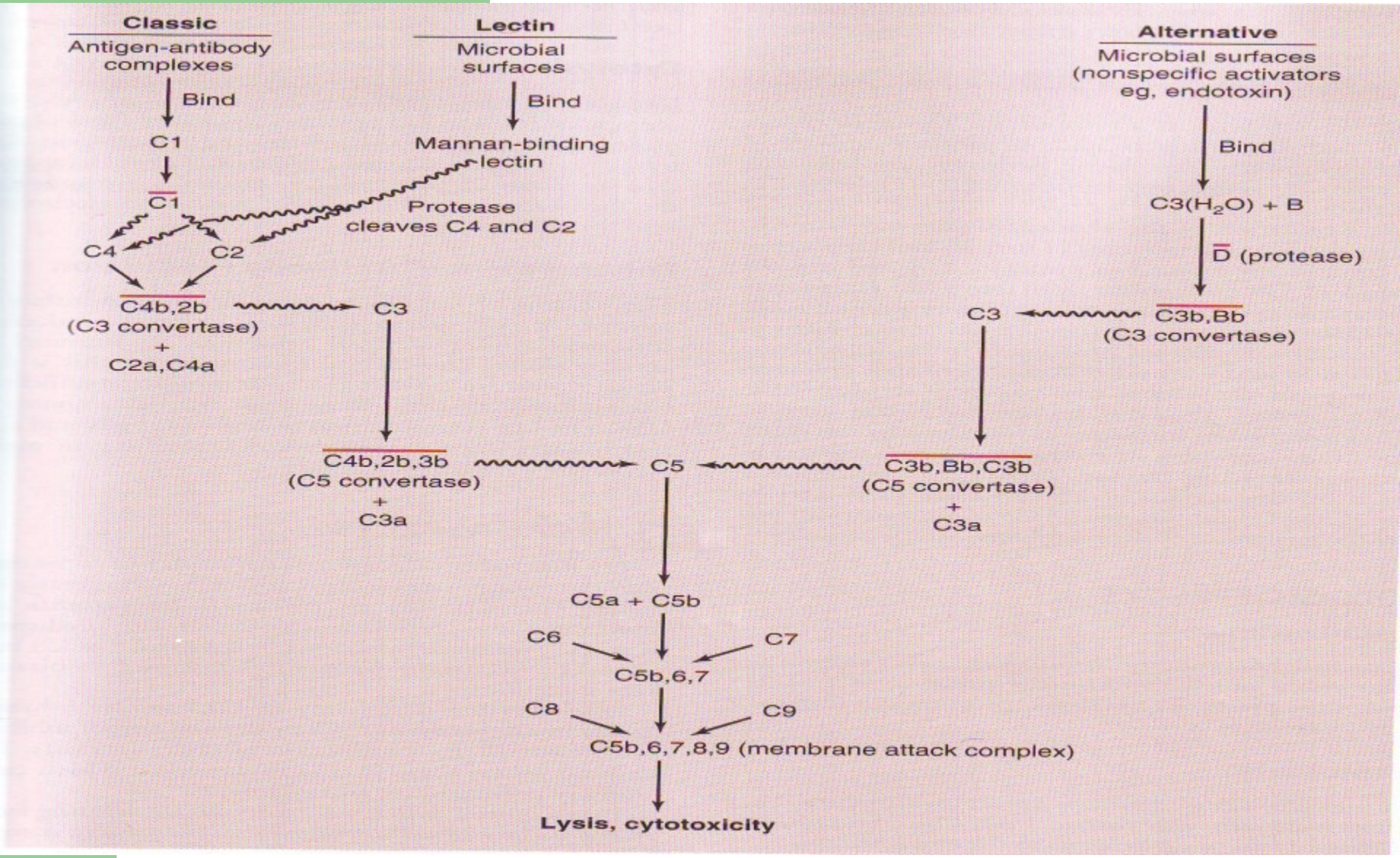
3- Lectin pathway:-

- In this pathway **MBL** also known as mannose-binding protein binds to the surface of microbes **bearing mannan**.
- This activates proteases associated with MBL that cleave C2 and C4 components of complement.
- Further, like the classical pathway, the C4b2a3b complex is formed, with C5 convertase activity resulting in the formation of C5b.
- C5b attaches to C6 and initiates the formation of the membrane attack complex.

- After MBL binds to the surface of a pathogen, MBL associated serine proteases bind to mannose-binding lectin. These MBL associated serine proteases are MASP-1 and MASP-2.







(Levenson, 2006)

Biologic effects of complement components:-

- **1-Opsonization:-**

  This occurs when activated complement component **coat pathogenic organisms** or immune complexes facilitating the process of phagocytosis.
- **2-Chemotaxis:-**
- C5a and the C567 complex **attract neutrophils**. C5a also enhances the **adhesiveness** of neutrophils to the endothelium.

Biologic effects of complement components:-

- **3-Anaphylotoxin:-**

  C3a,C4a and C5a cause **degranulation of mast cells** with release of mediators as **histamine** leading to increased vascular permeability and smooth muscle contraction.

Biologic effects of complement components:-

- **4-Cytolysis:-**

  In the final stage of the complement cascade, the membrane of target cells(erythrocytes, bacteria and tumor cells) are attacked by **membrane attack complex(MAC)** leading to cell destruction by entry of water and electrolytes into the cell.

Principle inflammatory activities of activated

Component	Activity
C3a, C5a	Anaphylatoxin—releases histamine, and other vasoactive compounds from basophils and mast cells, increasing capillary permeability
C3b, iC3b, C4b	Immune adherence and opsonization—binds antigen—antibody complexes to membranes of macrophages and neutrophils, enhancing phagocytosis; also binds complexes to erythrocytes, facilitating removal by the liver and spleen
C5a	Chemotaxis and chemokinesis—attracts phagocytic cells to sites of inflammation and increases their overall activity
C8, C9	Membrane damage—transmembrane channels form, permitting flux of cytoplasmic constituents. Mammalian cells swell and burst; bacterial cells become leaky and lose vital intracellular metabolites, but usually do not burst
Ba	Neutrophil chemotaxis
Bb	Macrophage activation—causes macrophages to adhere to surfaces and spread on them

(Hyde, 2000)

Deficiency of the complement components causes many diseases as follow:-

Deficiency of the complement components	Diseases
C1	• Autoimmune diseases, e.g. SLE • Increased susceptibility to infections • Nephritis
C4 , C2	• SLE
C3	• Recurrent bacterial infections • Chronic glomerulonephritis
C5	• Recurrent infections with Neisseria • SLE
C6	• Recurrent infections with Neisseria, Raynauds phenomenon
C7	• Recurrent infections with Neisseria • Raynauds phenomenon • Glomerulonephritis
C8	• Recurrent infections with Neisseria • Raynauds phenomenon • SLE
C1 esterase inhibitor	• Hereditary angioneurotic oedema
Properdin	• Severe meningococcal infections

(Hyde, 2000)

Summary

The complement system consists of a series of proteins that work to "complement" the work of antibodies in destroying microorganism.

There are three pathways for activation of complement cascade, classical , alternative and lectin pathways. The system is undergo three phases Initiation phase, amplification phase and membrane attack phase.

References

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Thanks a lot